COMSATS University, Attock Campus
Department of Computer Science
Topics in AI (Assignment No. I/CLO-I)

Due Date: 20/10/2026. **Points:** 25 + 25 =50

Submission Mode: GitHub Repository Link + Technical Report (PDF)

Assignment Title: Custom Reinforcement Learning Agent Implementation using Farama Gymnasium.

Objective: This assignment aims to provide hands-on experience in implementing, training, and evaluating a Reinforcement Learning (RL) algorithm from scratch using Python and the `gymnasium` library. You will apply an RL algorithm to either a standard benchmark toy problem or a real-world decision-making environment, analyze convergence, and evaluate hyperparameter sensitivity.

Task:

1. Select one environment type from Recommended Environments: `FrozenLake-v1`, `Taxi-v3`, or `CliffWalking-v1` `CartPole-v1`, `LunarLander-v2`, `MountainCar-v0`, or a custom Gymnasium environment (e.g., Grid-world Traffic Management, Portfolio Allocation).
2. Initialize your chosen environment using `gymnasium.make()`.
3. Implement the your chosen RL algorithm
4. Render and save a short GIF or video clip showing your trained agent successfully navigating/solving the environment.

Submission Deliverables

1. GitHub Repository:

Fully functional Python source code (`main.py`, `agent.py`, `utils.py`, `requirements.txt`).
Clean Jupyter Notebook (`.ipynb`) with executed cells displaying training progress and visualizations.
Saved trained model weights/Q-table (`.npy` or `.pt`/`.h5`).
Rendered GIF/MP4 showing the agent executing the learned policy.

2. Technical Report (PDF - Max 4 Pages):

Problem Description: Overview of chosen environment, state/action representation, and reward structure.
Methodology: Mathematical formulation of the algorithm and implementation architecture.
Experimental Setup & Hyperparameters: Table listing final tuned hyperparameters.
Results & Discussion: Plots, comparative analysis, and discussion of failure modes or key observations.
Conclusion: Summary of findings.